Technical Case Study: AI-Powered Talent Acquisition & Recruitment Pipeline

Stack: n8n, Google Gemini, Airtable, JavaScript, Gmail API

1. Executive Summary

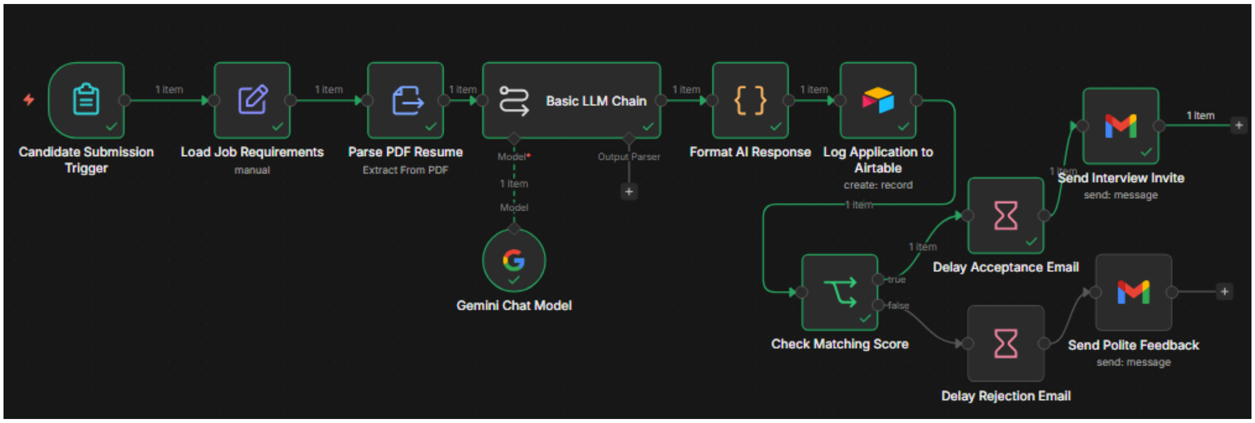
This project addresses the inefficiencies in modern recruitment by automating the end-to-end processing of job applications. By integrating Large Language Models (LLMs) with workflow automation, the system extracts data from resumes, evaluates candidate suitability against specific job descriptions, and manages professional communication without manual oversight.

2. System Architecture & Workflow

The pipeline is orchestrated using **n8n**, connecting several decoupled services into a single autonomous stream.

Workflow Stages:

1. **Trigger:** A "Job Application Form" node receives the candidate's name, email, and resume.
2. **Data Extraction:** The "Extract from File" node parses the raw text from the uploaded PDF resume.
3. **AI Analysis:** The "Basic LLM Chain" (powered by Google Gemini) compares the resume text against a provided Job Description.
4. **Data Transformation:** A custom JavaScript "Code" node cleans the AI's output, removing Markdown artifacts to ensure valid JSON formatting.
5. **Database Integration:** Candidate details, including a match score and AI-generated reasoning, are logged into an Airtable base.
6. **Logic Branching:** An "IF" node directs the flow based on a threshold score of 70.
7. **Automated Communication:** Based on the score, the system pauses via a "Wait" node before sending a personalized HTML email via the Gmail API.



3. Implementation Deep-Dive

A. Prompt Engineering & Scoring

The system utilizes a structured system prompt to force the LLM to output a precise JSON schema. It evaluates the candidate on a 0–100 scale and generates a concise, 2-sentence feedback response.

Status Logic:

- **Score ≥ 70:** Status set to "Shortlisted".
- **Score < 70:** Status set to "Rejected".

B. Technical Challenges: Data Sanitization

A common challenge with LLMs is the inclusion of Markdown formatting (backticks) in JSON responses. To maintain a robust pipeline, I authored a JavaScript transformation to sanitize the data:

JavaScript Code:

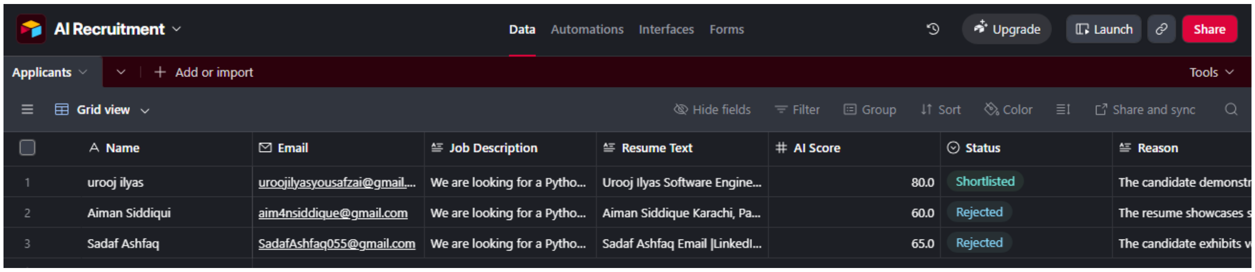
```
const rawText = $json.text.replace(/```json|```/g, "").trim();

return JSON.parse(rawText);
```

4. Results & Output

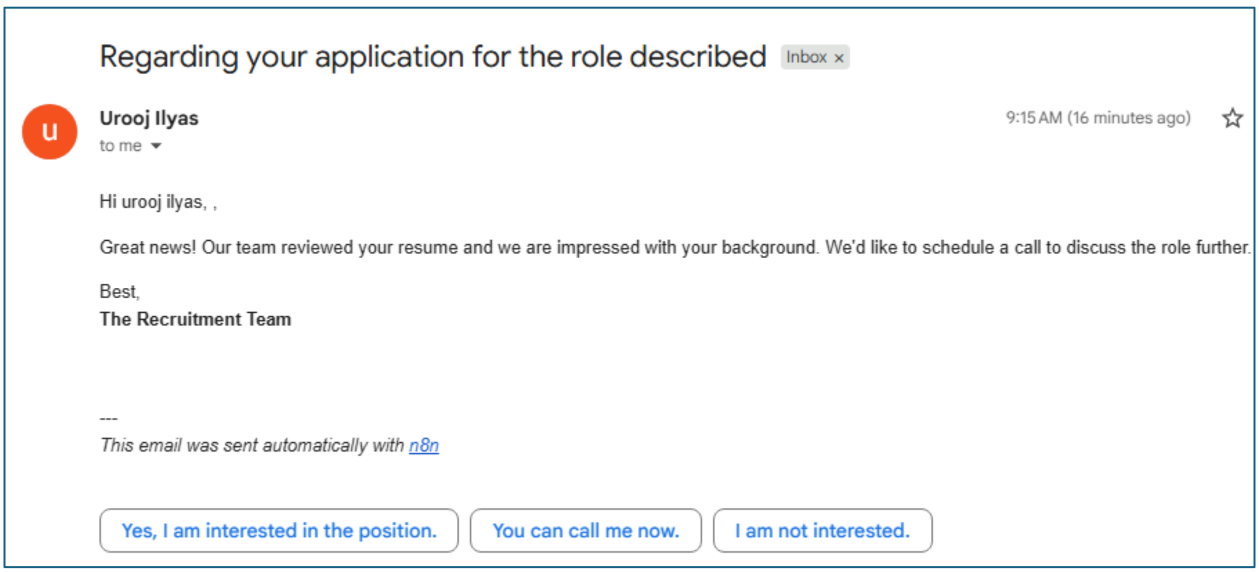
Data Logging (Airtable)

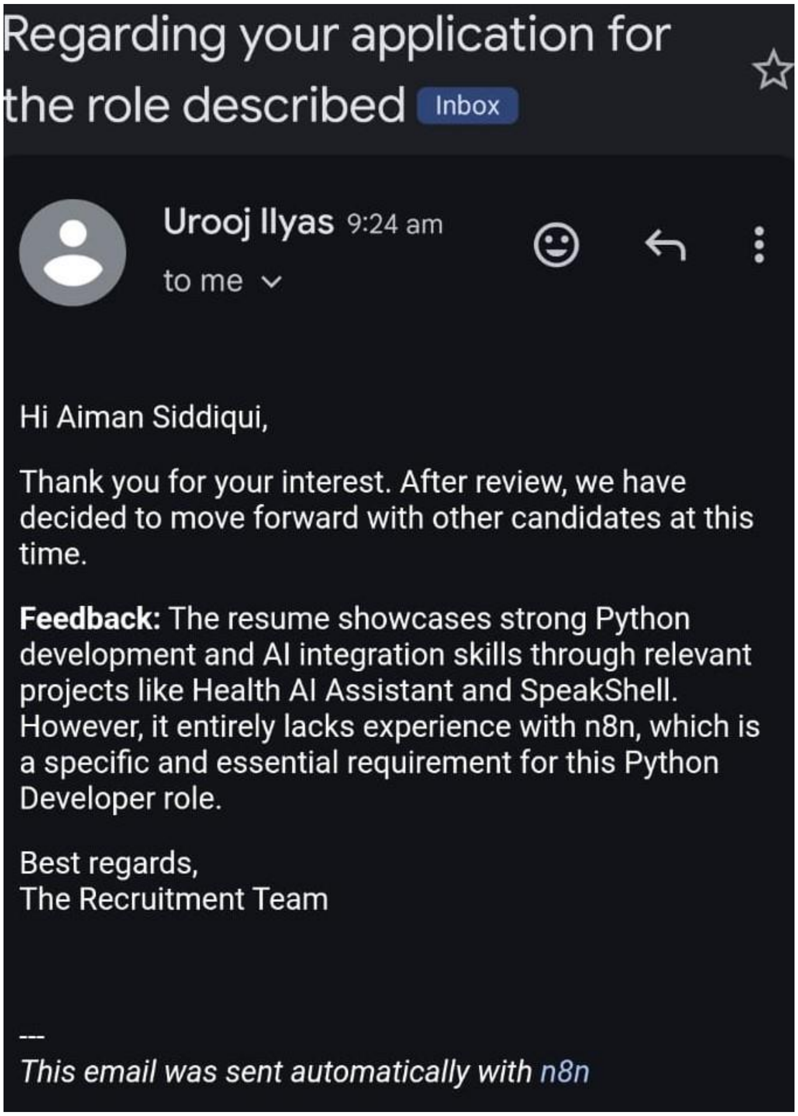
The system maintains a high-integrity database. Each record includes the candidate’s contact info, their AI-calculated score, and the qualitative "Reason" for the decision.



Candidate Experience (Gmail)

Candidates receive professionally formatted HTML emails. Rejections include specific feedback on missing skills, while accepted candidates receive an invitation for an interview.





Regarding your application for the role described Inbox

Urooj Ilyas 9:24 am
to me

Hi Aiman Siddiqui,

Thank you for your interest. After review, we have decided to move forward with other candidates at this time.

Feedback: The resume showcases strong Python development and AI integration skills through relevant projects like Health AI Assistant and SpeakShell. However, it entirely lacks experience with n8n, which is a specific and essential requirement for this Python Developer role.

Best regards,
The Recruitment Team

This email was sent automatically with n8n

5. Conclusion

This project demonstrates the practical application of AI in streamlining business operations. It showcases proficiency in API integration (OAuth2), data transformation with JavaScript, and the orchestration of complex AI-driven logic to provide tangible business value.